

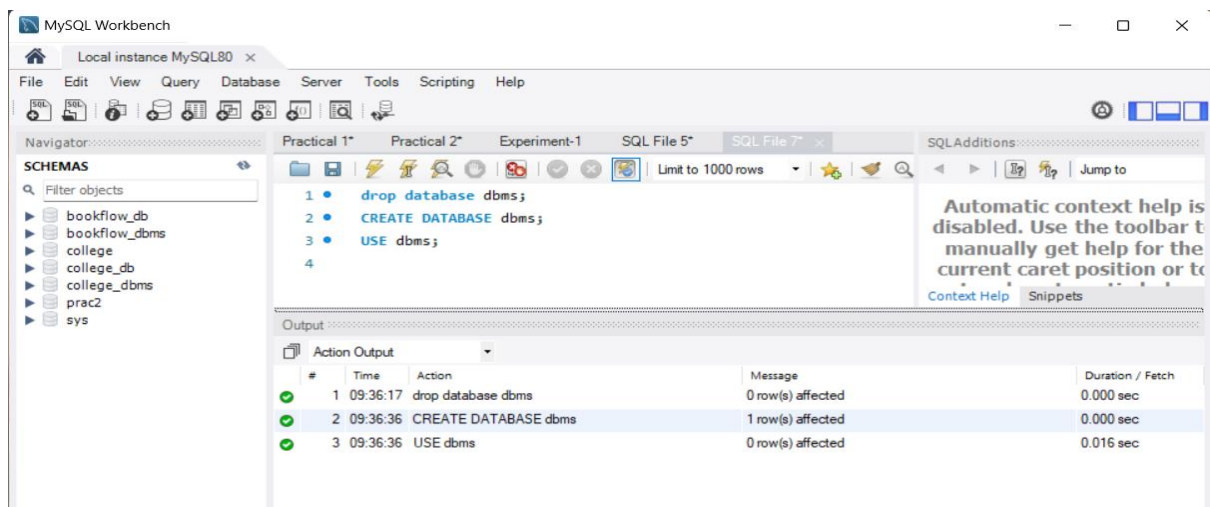
WEEK-1

Roll number: 2520090221

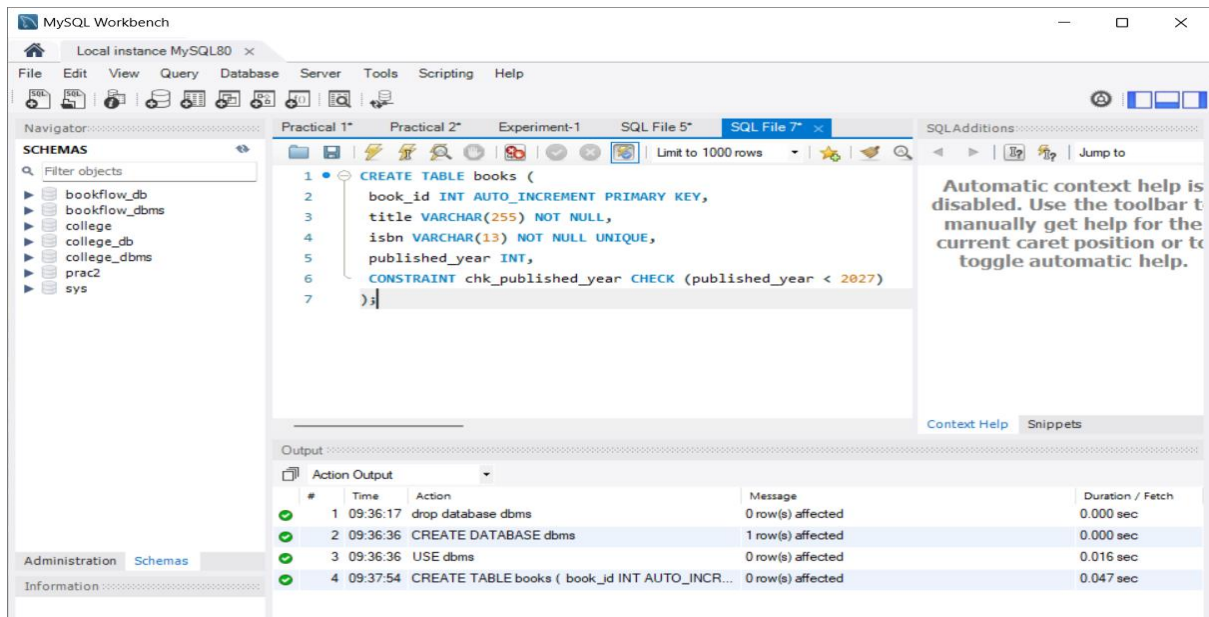
Name: D. VarunTeja Reddy

Q.)A local library is currently managing its books and members using a physical notebook, which has resulted in several operational problems such as misplaced books, difficulty tracking borrowed items, and inaccurate member records. To modernize its operations, the library plans to develop a database system called BookFlow. As a database developer, your task is to create the SQL foundation for this system. First, create a new database named bookflow_db. Then, design two tables: a Books table containing the columns book_id, title, isbn, and published_year, and a Members table containing the columns member_id, full_name, and email. Ensure data integrity by setting book_id and member_id as PRIMARY KEYS, applying a NOT NULL constraint to the title column so that every book must have a name, enforcing UNIQUE constraints on both isbn and email to prevent duplicate entries, and adding a CHECK constraint to ensure that the published_year is not in the future (for example, it must be less than 2027). Finally, insert three sample records into both the Books and Members tables to verify that the database structure and constraints are functioning correctly. This exercise will help you develop skills in schema definition, data integrity enforcement, and the use of SQL Data Definition Language (DDL) commands such as CREATE DATABASE, CREATE TABLE, and INSERT statements.

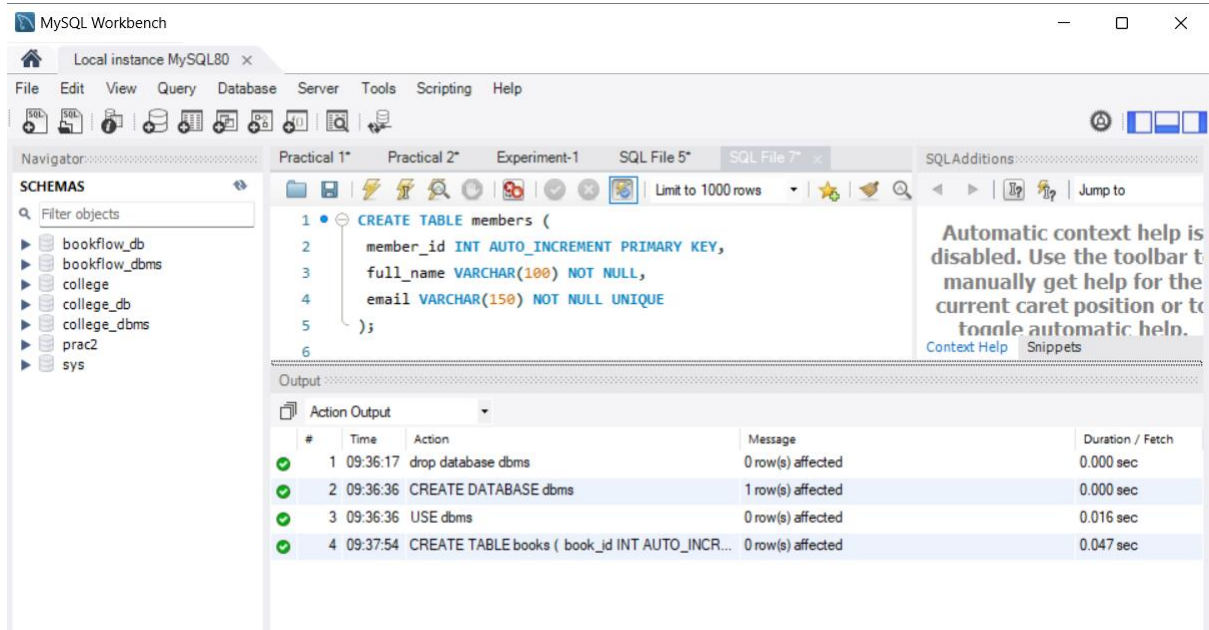
Step 1 — Create Database



Step 2 — Create Books Table (DDL + Constraints)



Query 4: Create the members table



Query 5 (verification): Inspect the table structures

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator: Schemas

Filter objects

- bookflow_db
- bookflow_dbms
- college
- college_db
- college_dbms
- prac2
- sys

Practical 1* Practical 2* Experiment-1 SQL File 5* SQL File 7* x SQLAdditions:

1 • DESCRIBE books;

2

Result Grid

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(255)	NO		NULL	
isbn	varchar(13)	NO	UNI	NULL	
published_year	int	YES		NULL	

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Form Editor

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	09:36:17	drop database dbms	0 row(s) affected	0.000 sec
2	09:36:36	CREATE DATABASE dbms	1 row(s) affected	0.000 sec
3	09:36:36	USE dbms	0 row(s) affected	0.016 sec
4	09:37:54	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.047 sec
5	09:40:47	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec

Administration Schemas

Information

No object selected

Limit to 1000 rows

1 • DESCRIBE members;

Result Grid

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_increment
full_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	18:06:17	drop database dbms	2 row(s) affected	0.110 sec
2	18:06:23	CREATE DATABASE dbms	1 row(s) affected	0.015 sec
3	18:06:23	USE dbms	0 row(s) affected	0.000 sec
4	18:06:28	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.078 sec
5	18:06:52	CREATE TABLE members (member_id INT AUTO...	0 row(s) affected	0.047 sec
6	18:06:56	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec
7	18:07:22	DESCRIBE members	3 row(s) returned	0.016 sec / 0.000 sec

Step 3 Initial Seeding (Valid Data)

Query 6: Insert 3 books

```

1  INSERT INTO books (title, isbn, published_year) VALUES
2  ('The Alchemist', '9780061122415', 1988),
3  ('Clean Code', '9780132350884', 2008),
4  ('Atomic Habits', '9780735211292', 2018);

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	18:06:17	drop database dbms	2 row(s) affected	0.110 sec
✓ 2	18:06:23	CREATE DATABASE dbms	1 row(s) affected	0.015 sec
✓ 3	18:06:23	USE dbms	0 row(s) affected	0.000 sec
✓ 4	18:06:28	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.078 sec
✓ 5	18:06:52	CREATE TABLE members (member_id INT AUTO...	0 row(s) affected	0.047 sec
✓ 6	18:06:56	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec
✓ 7	18:07:22	DESCRIBE members	3 row(s) returned	0.016 sec / 0.000 sec
✓ 8	18:09:01	INSERT INTO books (title, isbn, published_year) VA...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.000 sec

Query 7: Insert 3 members

```

1  INSERT INTO members (full_name, email) VALUES
2  ('Anil Kumar', 'anil.kumar@example.com'),
3  ('Priya Sharma', 'priya.sharma@example.com'),
4  ('Ravi Verma', 'ravi.verma@example.com');

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	18:06:17	drop database dbms	2 row(s) affected	0.110 sec
✓ 2	18:06:23	CREATE DATABASE dbms	1 row(s) affected	0.015 sec
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✓ 4	18:06:28	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.078 sec
✓ 5	18:06:52	CREATE TABLE members (member_id INT AUTO...	0 row(s) affected	0.047 sec
✓ 6	18:06:56	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec
✓ 7	18:07:22	DESCRIBE members	3 row(s) returned	0.016 sec / 0.000 sec
✓ 8	18:09:01	INSERT INTO books (title, isbn, published_year) VA...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.000 sec
✓ 9	18:10:09	INSERT INTO members (full_name, email) VALUES...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.000 sec

OUTPUT:

1 • `SELECT * FROM books;`

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018
NULL	NULL	NULL	NULL

books 2 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	18:06:17	drop database dbms	2 row(s) affected	0.110 sec
✓ 2	18:06:23	CREATE DATABASE dbms	1 row(s) affected	0.015 sec
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✓ 4	18:06:28	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.078 sec
✓ 5	18:06:52	CREATE TABLE members (member_id INT AUTO...	0 row(s) affected	0.047 sec
✓ 6	18:06:56	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec
✓ 7	18:07:22	DESCRIBE members	3 row(s) returned	0.016 sec / 0.000 sec
✓ 8	18:09:01	INSERT INTO books (title, isbn, published_year) VA...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.000 sec
✓ 9	18:10:09	INSERT INTO members (full_name, email) VALUES...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.000 sec
✓ 10	18:10:42	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

1 • `SELECT * FROM members;`

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

member_id	full_name	email
1	AI Refresh data re-executing the original query	
2	Priya Sharma	priya.sharma@example.com
3	Ravi Verma	ravi.verma@example.com
NULL	NULL	NULL

members 3 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	18:06:17	drop database dbms	2 row(s) affected	0.110 sec
✓ 2	18:06:23	CREATE DATABASE dbms	1 row(s) affected	0.015 sec
✓ 3	18:06:23	USE dbms	0 row(s) affected	0.000 sec
✓ 4	18:06:28	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.078 sec
✓ 5	18:06:52	CREATE TABLE members (member_id INT AUTO...	0 row(s) affected	0.047 sec
✓ 6	18:06:56	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec
✓ 7	18:07:22	DESCRIBE members	3 row(s) returned	0.016 sec / 0.000 sec
✓ 8	18:09:01	INSERT INTO books (title, isbn, published_year) VA...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.000 sec
✓ 9	18:10:09	INSERT INTO members (full_name, email) VALUES...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.000 sec
✓ 10	18:10:42	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 11	18:40:57	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec

Step: 4 Constraint Tests

Test 1: Duplicate ISBN (UNIQUE violation)

Limit to 1000 rows

```

1 INSERT INTO books (title, isbn, published_year)
2 VALUES ('Fake Copy', '9780061122415', 2000);

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	18:06:17	drop database dbms	2 row(s) affected	0.110 sec
2	18:06:23	CREATE DATABASE dbms	1 row(s) affected	0.015 sec
3	18:06:23	USE dbms	0 row(s) affected	0.000 sec
4	18:06:28	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.078 sec
5	18:06:52	CREATE TABLE members (member_id INT AUTO...	0 row(s) affected	0.047 sec
6	18:06:56	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec
7	18:07:22	DESCRIBE members	3 row(s) returned	0.016 sec / 0.000 sec
8	18:09:01	INSERT INTO books (title, isbn, published_year) VA...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.000 sec
9	18:10:09	INSERT INTO members (full_name, email) VALUES...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.000 sec
10	18:10:42	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
11	18:40:57	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec
12	18:42:02	INSERT INTO books (title, isbn, published_year) VA...	Error Code: 1062. Duplicate entry '9780061122415' ...	0.016 sec

Test 2: NULL title:

Limit to 1000 rows

```

1 INSERT INTO books (title, isbn, published_year)
2 VALUES (NULL, '9999999999999', 2010);

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	18:06:17	drop database dbms	2 row(s) affected	0.110 sec
2	18:06:23	CREATE DATABASE dbms	1 row(s) affected	0.015 sec
3	18:06:23	USE dbms	0 row(s) affected	0.000 sec
4	18:06:28	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.078 sec
5	18:06:52	CREATE TABLE members (member_id INT AUTO...	0 row(s) affected	0.047 sec
6	18:06:56	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec
7	18:07:22	DESCRIBE members	3 row(s) returned	0.016 sec / 0.000 sec
8	18:09:01	INSERT INTO books (title, isbn, published_year) VA...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.000 sec
9	18:10:09	INSERT INTO members (full_name, email) VALUES...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.000 sec
10	18:10:42	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
11	18:40:57	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec
12	18:42:02	INSERT INTO books (title, isbn, published_year) VA...	Error Code: 1062. Duplicate entry '9780061122415' ...	0.016 sec
13	18:42:44	INSERT INTO books (title, isbn, published_year) VA...	Error Code: 1048. Column 'title' cannot be null	0.000 sec

Test 3: Future publication year

1 • `INSERT INTO books (title, isbn, published_year)`
2 `VALUES ('Time Traveler', '888888888888', 2030);`

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	18:06:17	drop database dbms	2 row(s) affected	0.110 sec
✓ 2	18:06:23	CREATE DATABASE dbms	1 row(s) affected	0.015 sec
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✓ 4	18:06:28	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.078 sec
✓ 5	18:06:52	CREATE TABLE members (member_id INT AUTO...	0 row(s) affected	0.047 sec
✓ 6	18:06:56	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec
✓ 7	18:07:22	DESCRIBE members	3 row(s) returned	0.016 sec / 0.000 sec
✓ 8	18:09:01	INSERT INTO books (title, isbn, published_year) VA...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.000 sec
✓ 9	18:10:09	INSERT INTO members (full_name, email) VALUES...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.000 sec
✓ 10	18:10:42	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 11	18:40:57	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec
✗ 12	18:42:02	INSERT INTO books (title, isbn, published_year) VA...	Error Code: 1062. Duplicate entry '9780061122415' ...	0.016 sec
✗ 13	18:42:44	INSERT INTO books (title, isbn, published_year) VA...	Error Code: 1048. Column 'title' cannot be null	0.000 sec
✗ 14	18:43:11	INSERT INTO books (title, isbn, published_year) VA...	Error Code: 3819. Check constraint 'chk_published...	0.000 sec

Test 4: Duplicate email

1 • `INSERT INTO members (full_name, email)`
2 `VALUES ('Anil Clone', 'anil.kumar@example.com');`

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	18:06:17	drop database dbms	2 row(s) affected	0.110 sec
✓ 2	18:06:23	CREATE DATABASE dbms	1 row(s) affected	0.015 sec
✓ 3	18:06:23	USE dbms	0 row(s) affected	0.000 sec
✓ 4	18:06:28	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.078 sec
✓ 5	18:06:52	CREATE TABLE members (member_id INT AUTO...	0 row(s) affected	0.047 sec
✓ 6	18:06:56	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec
✓ 7	18:07:22	DESCRIBE members	3 row(s) returned	0.016 sec / 0.000 sec
✓ 8	18:09:01	INSERT INTO books (title, isbn, published_year) VA...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.000 sec
✓ 9	18:10:09	INSERT INTO members (full_name, email) VALUES...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.000 sec
✓ 10	18:10:42	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 11	18:40:57	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec
✗ 12	18:42:02	INSERT INTO books (title, isbn, published_year) VA...	Error Code: 1062. Duplicate entry '9780061122415' ...	0.016 sec
✗ 13	18:42:44	INSERT INTO books (title, isbn, published_year) VA...	Error Code: 1048. Column 'title' cannot be null	0.000 sec
✗ 14	18:43:11	INSERT INTO books (title, isbn, published_year) VA...	Error Code: 3819. Check constraint 'chk_published...	0.000 sec
✗ 15	18:43:56	INSERT INTO members (full_name, email) VALUES...	Error Code: 1062. Duplicate entry 'anil.kumar@exa...	0.000 sec

